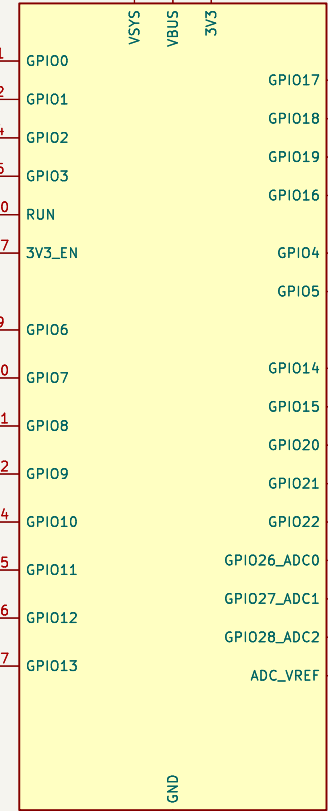


Controller and input voltages

MEMRISTOR CNN | 4 inputs x 2 outputs | Hardware prototype – Rev B

USB to laptop / Python

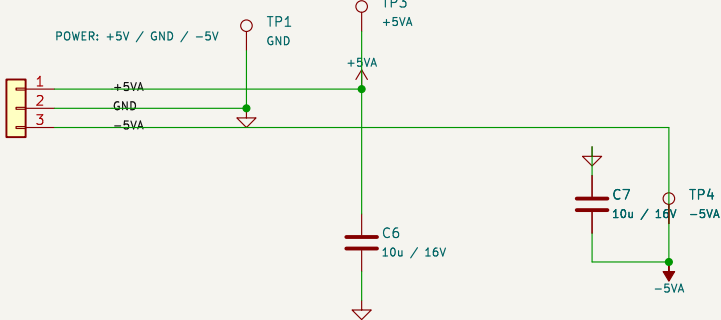
A1
RaspberryPi Pico



Read mode at reset

External analogue supply

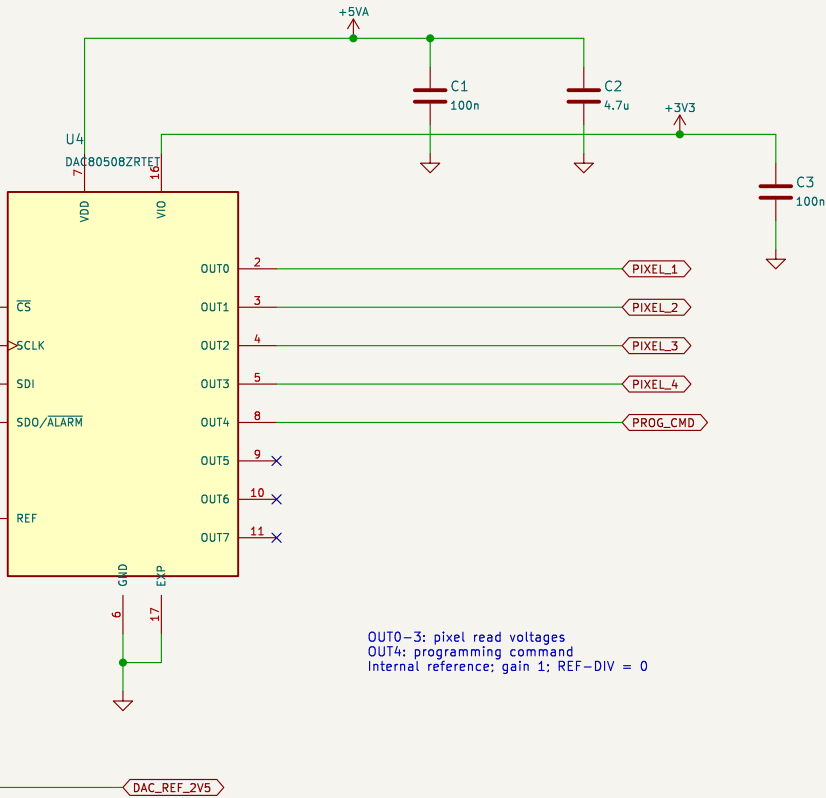
J1



Pico is USB powered. J1 is a separate regulated +/-5 V supply.
Power USB first; remove analogue rails before unplugging USB.

+5VA

-5VA



OUT0–3: pixel read voltages
OUT4: programming command
Internal reference; gain 1; REF–DIV = 0

Cell array

File: array.kicad_sch

Programming and readout

File: analog.kicad_sch

Prototype: external cells; device-specific validation required.

Sebish Niraula

Sheet: /
File: memristor_hardware.kicad_sch

Title: Memristor CNN – Controller and converters

Size: A3 | Date: 2026–10–05
KiCad E.D.A. | kicad–cli 7.0.11+dfsg–1build4

Rev: B
Id: 1/3

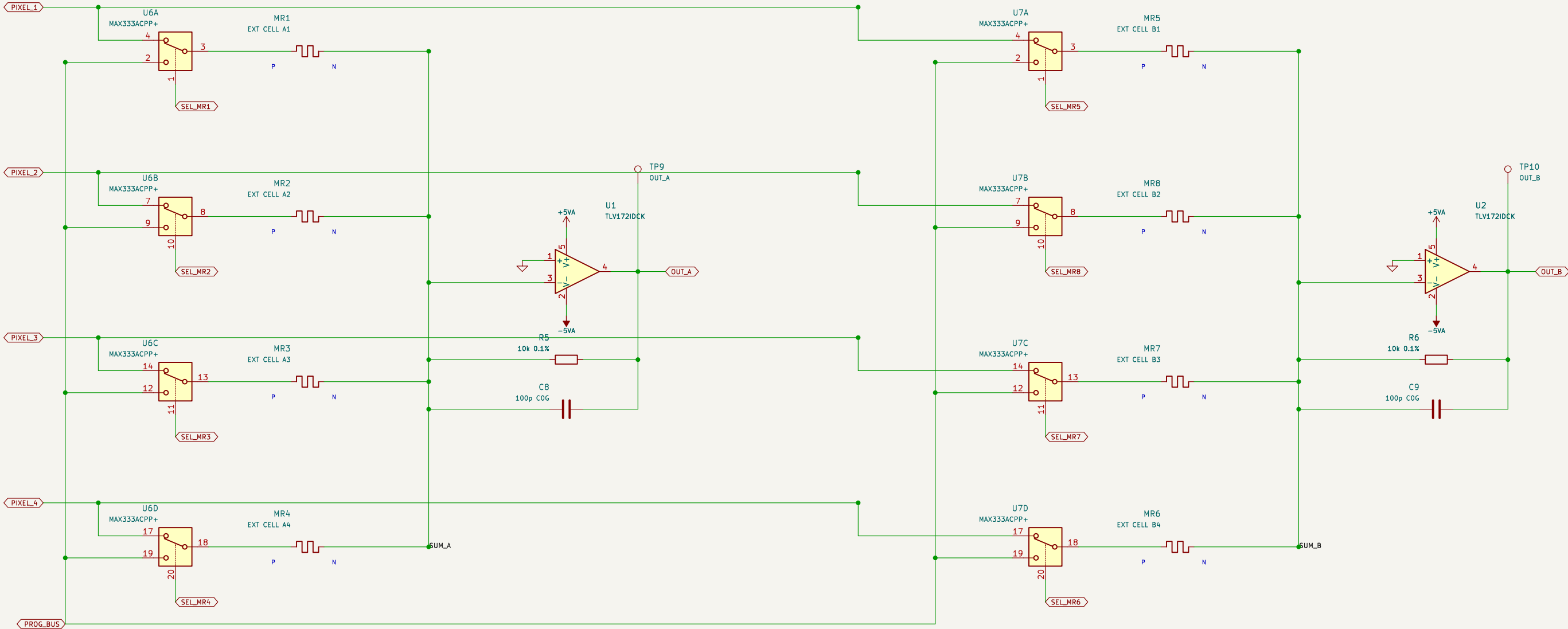
Cell array and current summing

MEMRISTOR CNN | 4 inputs x 2 outputs | Hardware prototype – Rev B

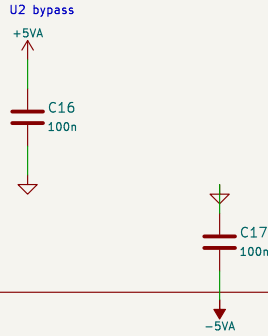
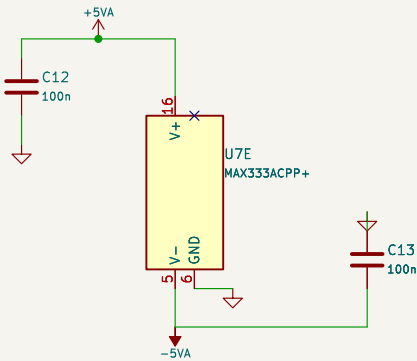
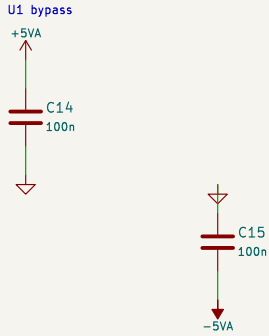
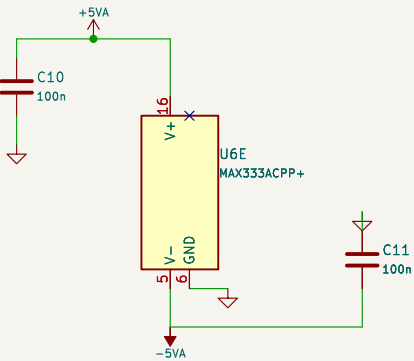
Switch LOW = read. Switch HIGH = programme one cell.

Filter A

Filter B



Crossing wires connect only at a junction dot.



MR1–MR8 are external–cell sockets (pin 1=P, pin 2=N).
Start with fixed resistors; the physical device has not yet been selected.

Prototype: external cells; device–specific validation required.

Sebish Niraula

Sheet: /Cell array/

File: array.kicad_sch

Title: Memristor CNN – Switched cell array

Size: A3 Date: 2026–10–05

KiCad E.D.A. kicad–cli 7.0.11+dfsg–1build4

Rev: B

Id: 2/3

MEMRISTOR CNN | 4 inputs x 2 outputs | Hardware prototype – Rev B

